

Podcast 15 — Serverless & Managed Apps

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Length target: 15 min **Repo references:** `domain-2-new-solutions/04-lambda.md` , `06-aws-batch.md` , `37-other-services.md` (App Runner, SAM, Beanstalk), `42-lightsail.md` , `39-proton.md`

Topic & Scope

Lambda is heavily tested; the surrounding “managed app” tier (App Runner, Beanstalk, Lightsail, Proton, Batch) creates distractors. Sort them.

Service Coverage Depth

Deep: - Lambda — runtime, layers, concurrency, provisioned concurrency, VPC config, destinations, event sources - App Runner vs Elastic Beanstalk vs Lightsail decision

Brief: - AWS Batch (managed batch with job queues) - AWS SAM (Lambda IaC) - AWS Proton (platform engineering)

Structured Outline

1. **Open (30s)** — “Lambda gets the most love. But App Runner, Beanstalk, Lightsail, Batch all appear as distractors. Let’s draw the lines.”
2. **Lambda Deep (5 min)** — execution model, runtimes, layers, environment variables, VPC config (ENI), reserved + provisioned concurrency, destinations, event source mappings (SQS, Kinesis, DynamoDB Streams), error handling
3. **Event Sources & Patterns (2 min)** — API Gateway, EventBridge, SQS, S3, DynamoDB Streams, Kinesis, ALB
4. **App Runner (2 min)** — managed containerized HTTP service, auto-deploys from GitHub/ECR, scales to zero
5. **Elastic Beanstalk (1.5 min)** — opinionated PaaS for EC2-based web apps, multiple deployment modes
6. **Lightsail + Batch + Proton (2 min)** — Lightsail = simple VPS distractor; Batch = job queues; Proton = platform engineering templates
7. **Decision Matrix + Rapid-Fire (2 min)**

Must-Mention Exam Tips

- Lambda max: 15-min timeout, 10 GB memory, 10 GB ephemeral /tmp, 250 MB unzipped package (or 10 GB container image)

- Lambda concurrency: account-level (1,000 default), reserved (cap per function), provisioned (pre-warmed)
- Lambda cold start mitigation: provisioned concurrency, smaller package, SnapStart for Java
- Lambda in VPC: ENI created in subnets, can talk to RDS / ElastiCache in VPC; no internet unless NAT GW
- Event source mappings (poll-based): SQS, Kinesis, DynamoDB Streams, MSK, MQ, Self-managed Kafka
- Async invocation: destinations for success/failure (SNS, SQS, Lambda, EventBridge); DLQ for fallback
- Lambda Layers: shared dependencies up to 5 layers per function
- App Runner: managed container web service, automatic scaling + HTTPS, scales to zero
- Elastic Beanstalk: managed EC2-based PaaS, supports Docker / Java / .NET / Node / Python / Ruby
- AWS Batch: managed job orchestration on EC2 / Fargate / Spot with job queues + compute environments
- AWS SAM: Lambda-focused IaC (extends CloudFormation)
- Proton: platform engineering tool — environment + service templates for dev teams

Must-Mention Exam Traps

- EXAM TRAP: Lambda max execution is 15 minutes — for longer, use Step Functions / ECS / EC2 / Batch
- EXAM TRAP: Lambda in VPC adds cold-start latency (ENI provisioning) — improved with Hyperplane ENI
- EXAM TRAP: Provisioned concurrency costs money even when not used — only for latency-sensitive paths
- EXAM TRAP: Lambda reserved concurrency CAPS — limits how much that function can scale
- EXAM TRAP: SQS DLQ on Lambda is for failures the function couldn't process — not for batch DLQ
- EXAM TRAP: App Runner ≠ Elastic Beanstalk — App Runner is container; Beanstalk is EC2-based
- EXAM TRAP: Lightsail is rarely the right answer in SAP-C02 — usually distractor for “predictable cheap”
- EXAM TRAP: Batch ≠ Step Functions — Batch is job queues at scale; Step Functions is workflow orchestration
- EXAM TRAP: Beanstalk creates resources you might not see — uses CloudFormation under the hood
- EXAM TRAP: Proton ≠ Service Catalog — Proton is continuous platform tooling; Catalog is one-shot products
- EXAM TRAP: Lambda function URLs are a feature for HTTPS endpoint — not always best vs API Gateway (no auth/throttling/caching features)
- EXAM TRAP: SAM ≠ CDK — both are IaC; SAM is Lambda-focused YAML, CDK is general-purpose code

- EXAM TRAP: Lambda response size max 6 MB sync, 256 KB async invocation payload

Key Decision Matrix

Scenario	Pick
Event-driven short function	Lambda
Long-running job > 15 min	ECS / Fargate / Batch / EC2
Containerized web service, auto-scale to zero	App Runner
EC2-based managed PaaS, multiple languages	Elastic Beanstalk
Massive batch jobs with queue/priority	AWS Batch
Personal blog / WordPress on AWS	Lightsail
Platform engineering templates for many teams	Proton
Reduce Lambda cold starts on critical path	Provisioned Concurrency / SnapStart
Lambda access to RDS in private subnet	Lambda in VPC + ENI

Tone & Style

- Lead with Lambda — most exam questions live there
- Frame App Runner as “Lambda for containers (HTTP only)”
- Lightsail mentioned as the trap distractor

Rapid-Fire Closer

“Lambda max runtime?” — “15 min.” “App Runner workload?” — “Container HTTP.” “Beanstalk underlying?” — “EC2 PaaS.” “Lightsail right answer?” — “RARELY.” “Lambda in VPC needs internet?” — “NAT Gateway.”